RÉPUBLIQUE FRANÇAISE
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AI for Research & Science

Three use-cases

Presentation at the NEO-AMELEAS Workshop

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01

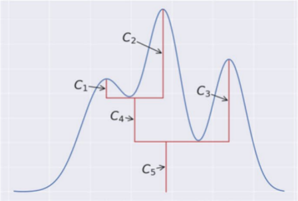
AI for Skeptical People (I)

Creating Vector Images with TikZ

Enhancing Visual Quality: Fast Vector Graphics with AI

- **Fast & Professional Figures**: AI allows you to generate high-quality illustrations quickly without wasting time.
- **What is TikZ?**: A native LaTeX programming language for vector graphics that integrates perfectly with your document.
- **Full Control**: Since TikZ figures are code-based, you always keep full control to manually edit or fine-tune any detail at any time.

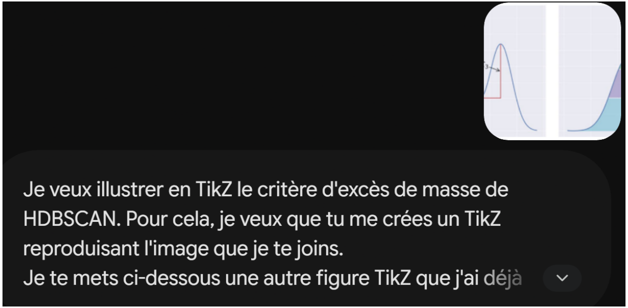
Example: Target Raster Image



The Challenge:

- A 1D density plot illustrating the *excess of mass* in HDBSCAN.
- Simple raster formats (PNG/JPG) are pixelated and cannot match document fonts.

Example: Prompting the AI



Human-AI Collaboration:

- Simply upload the target image and ask the AI to reproduce it in TikZ.
- Saves hours of calculating coordinates and writing syntax manually.

Example: The Generated TikZ Code

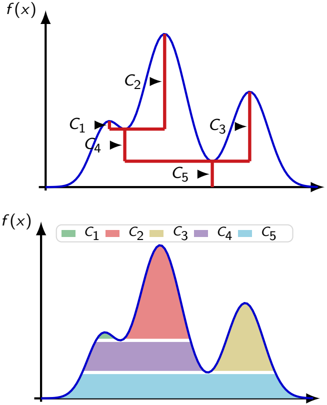
TikZ Code Structure:

```
\begin{tikzpicture}[scale=0.45]
  \tikzset{declare function={f(x) = ...}}

  % Draw density curve
  \draw[thick, blue] plot[domain=0:8] ...;

  % Draw cluster tree bars
  \draw[very thick, red] (1.87,1.70) -- ...;

  % Add labels
  \draw[<-, semithick] (1.87,1.82) -- ...;

\end{tikzpicture}
```

LaTeX Integration:

- **Include Package**:
  - Add `\usepackage{tikz}` in your preamble.
- **Direct Placement**:
  - Paste the generated code directly in the document body.
- **No Assets to Manage**:
  - No need for external image files (PNG/PDF).

Example: The Native TikZ Result



Why it works:

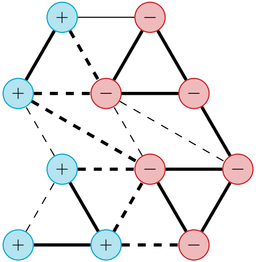
- Native vector rendering means infinite scaling and crispness.
- Complete control over font size and styling.

02

AI for Skeptical People (II)

Finding a Mathematical Counter-Example

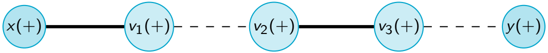
Case Study: GSBM on Lattices



GSBM on a Triangular Grid:

- **Ground Truth** $\Sigma_x \in \{+1, -1\}$:
  - Node colors represent spins (50% $+$, 50% $-$).
- **Observed signs** $Y_e \in \{+1, -1\}$:
  - $\mathbb{P}(Y_e = \Sigma_x \Sigma_y) = p = 0.75$.
  - **Graphical Code**:
    - **Solid lines**: Ferromagnetic ($Y_e = +1$).
    - **Dashed lines**: Antiferromagnetic ($Y_e = -1$).
    - **Bold lines**: Satisfied ($Y_e = \Sigma_x \Sigma_y$).
    - **Thin lines**: Frustrated ($Y_e \neq \Sigma_x \Sigma_y$).

Case Study: The Long-Path Intuition



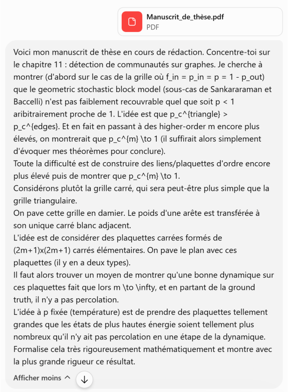
Why the Intuition Suggested No Recovery:

- For two distant nodes x, y (both $+ \in \Sigma$), the product of observed signs along a path of length n must be $+1$ for correct inference.
- This requires an **even** number of antiferromagnetic edges (each occurring with probability $q = 1 - p = 0.25$).
- The probability of having an odd number of antiferromagnetic edges is:

$$\mathbb{P}(\text{odd}) = \frac{1 - (2p-1)^n}{2} = \frac{1 - (0.5)^n}{2} \xrightarrow[n\to\infty]{} \frac{1}{2}$$

- The path correlation vanishes, suggesting no weak recovery is possible for any $p < 1$.
- *Simulation Artifact*: Vanishing overlap on large grids was actually due to the MCMC dynamics' mixing time.

Case Study: Prompting the AI



Manuscrit_de_thèse.pdf
PDF

Voici mon manuscrit de thèse en cours de rédaction. Concentre-toi sur le chapitre 11 : détection de communautés sur graphes. Je cherche à montrer (d'abord sur le cas de la grille où f_in = p_in = p = 1 - p_out) que le geometric stochastic block model (sous-cas de Sankararaman et Baccelli) n'est pas faiblement recouvrable quel que soit p < 1 aribitrairement proche de 1. L'idée est que p_c^{triangle} > p_c^{edges}. Et en fait en passant à des higher-order m encore plus élevés, on montrerait que p_c^{m} \to 1 (il suffirait alors simplement d'évoquer mes théorèmes pour conclure).
Toute la difficulté est de construire des liens/plaquettes d'ordre encore plus élevé puis de montrer que p_c^{m} \to 1.
Considérons plutôt la grille carré, qui sera peut-être plus simple que la grille triangulaire.
On pave cette grille en damier. Le poids d'une arête est transférée à son unique carré blanc adjacent.
L'idée est de considérer des plaquettes carrées formés de (2m+1)x(2m+1) carrés élémentaires. On pave le plan avec ces plaquettes (il y en a deux types).
Il faut alors trouver un moyen de montrer qu'une bonne dynamique sur ces plaquettes fait que lors m \to \infty, et en partant de la ground truth, il n'y a pas percolation.
L'idée à p fixée (température) est de prendre des plaquettes tellement grandes que les états de plus hautes énergie soient tellement plus nombreux qu'il n'y ait pas percolation en une étape de la dynamique.
Formalise cela très rigoureusement mathématiquement et montre avec la plus grande rigueur ce résultat.

Afficher moins

The Research Question:

- We uploaded our draft manuscript describing a plaquette dynamics.
- We asked the AI to formalize the proof that the threshold $p_c^{(m)} \to 1$ as the plaquette size $m \to \infty$.
- If true, this would prove no recovery is possible for any $p < 1$.

Case Study: ChatGPT's Verdict

Réflexion durant 18m 5s ›

Il y a un problème assez net avec l'objectif annoncé : **sur la grille carrée, l'énoncé "non faiblement recouvrable pour tout $p < 1$" est faux**. Ce n'est pas une petite fissure technique dans le plâtre, c'est plutôt le mur porteur qui vient de déposer sa démission.

The AI's Answer:

- The model spent **18 minutes** reasoning.
- It immediately disproved the theorem: **"on the square lattice, the statement 'no weak recovery for all $p < 1$' is false."**
- Proving $p_c^{(m)} \to 1$ was impossible because the MAP estimator actually achieves recovery.

Case Study: Why MAP Works (Peierls Contours)

Proof Sketch by Peierls Argument:

- **Contours**: Boundaries of incorrect spin decisions form contours γ on the dual grid.
- **MAP Optimality**: Any contour γ in the MAP estimator must contain at least 50% corrupted edges (otherwise, flipping the region inside would improve the MAP score):

$$B(\gamma) \geq \frac{|\gamma|}{2}$$

- **Chernoff Bound**: The probability of a fixed contour γ of length ℓ being incorrect is bounded by $(2\sqrt{p(1-p)})^{\ell}$.
- **Contour Counting**: The number of contours of length ℓ around any node is at most $\ell 3^{\ell}$. The probability of an incorrect node is bounded by:

$$\sum_{\ell \geq 4} \ell (6\sqrt{p(1-p)})^{\ell} \xrightarrow[p \to 1]{} 0$$

- **Conclusion**: Weak recovery is achieved for $p > p_0$, saving months of trying to prove a false statement.

03

AI for Enthusiasts

Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)

Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)

Thank you!